

Medical OmniChannel Analytics

A unified, governed intelligence layer for Daiichi Sankyo oncology Medical Affairs — raw engagement signals flow through Lakeflow into one Unity Catalog 360° view, served operationally via Lakebase, made intelligent with Gen AI, queryable in Genie, and surfaced to MSLs in a Databricks App.

Lakeflow → Unity Catalog

Lakebase serving

Genie + AI/BI

Databricks App (Copilot)

One prepared, governed view of every KOL — ~\$3.4M/year of impact

360° KOL view

unified signal

MSL notes, inquiries, congress, digital, pubs & trials in one governed profile.

Prepared MSLs

minutes, not hours

AI pre-engagement briefing + next-best action for every KOL visit.

Governed & safe

audit-ready

Non-promotional, human-in-the-loop, one Unity Catalog security model.

~\$3.4M / yr

estimated value

For a 120-MSL field-medical org — KPI detail on the following slides.

For the sponsor and the field

Executive sponsor (VP Medical Affairs): quantified ROI, faster inquiry SLAs, and audit-ready Unity Catalog governance for a regulated, non-promotional function. Domain owner (field-medical lead): every MSL arrives prepared, prioritized, and armed with approved content.

Medical Affairs signal is real — but it's scattered

Daiichi Sankyo's MSLs engage the most influential oncology KOLs for Enhertu and Datroway — but the evidence of what a KOL actually cares about lives in disconnected systems, so follow-ups slip, medical inquiries age past SLA, and high-value KOLs go under-served.

Fragmented channels

MSL notes, medical inquiries, congress activity, digital engagement, publications & trials all live apart — no single HCP view.

Insight trapped in free text

The richest signal — scientific interest, unmet information needs, sentiment — sits in unstructured MSL interaction notes.

Reactive, manual prep

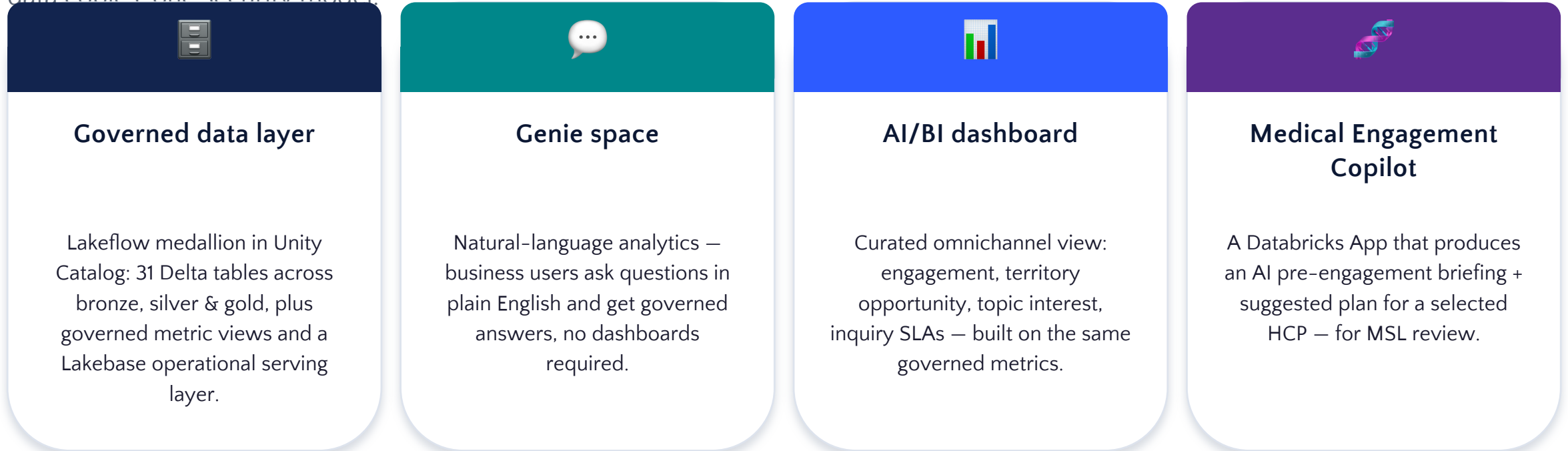
MSLs spend hours assembling context before each engagement; prioritization is gut-feel, not data-driven.

Governance is non-negotiable

Medical Affairs is non-promotional and regulated — every shared asset must be medically approved and human-reviewed.

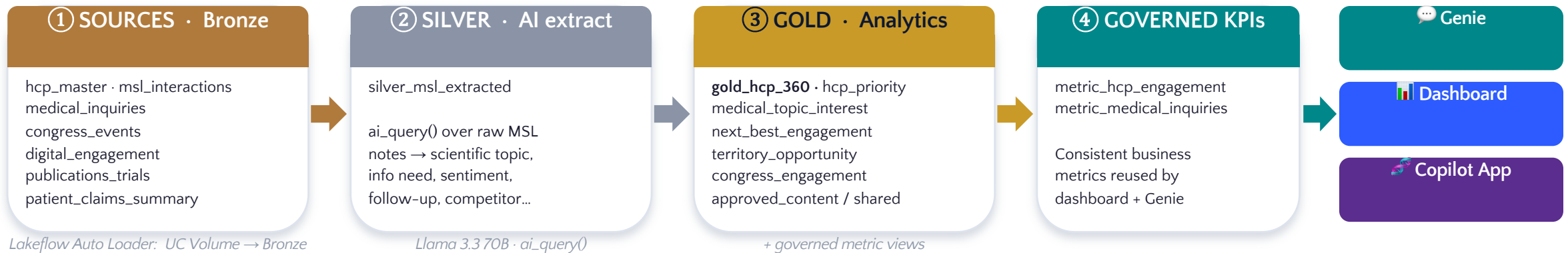
One governed platform – raw files to MSL action

Everything runs on the Databricks Data Intelligence Platform on one governed Unity Catalog schema (_sa701.moa). A Lakeflow medallion pipeline feeds Gold tables that power Genie, an AI/BI dashboard, a Lakebase-served MSL cockpit, and the Copilot app – no data copies, one security model.



Shared foundation: Lakeflow · Unity Catalog governance · Delta Lake · Lakebase (PostgreSQL) · Databricks-hosted LLM via ai_query() · serverless SQL

End-to-end data flow — source to decision



 **Databricks-hosted LLM** — databricks-meta-llama-3-3-70b-instruct · called via ai_query() for silver extraction AND the app's pre-engagement briefing

 **Unity Catalog** — single governance & lineage layer across every table, metric and surface · Delta Lake · Lakebase operational serving · serverless SQL

Genie, the AI/BI dashboard, the Lakebase MSL action queue, and the Copilot app all read the SAME governed Gold tables & metric views — one source of truth, one security model.

Lakeflow medallion in Unity Catalog (_sa701.moa)

BRONZE / SOURCE

- hcp_master
- msl_interactions (free-text notes)
- medical_inquiries
- congress_events
- digital_engagement
- publications_trials
- patient_claims_summary

Raw omnichannel signal landed as Delta tables

SILVER

- silver_msl_extracted

*Structured fields extracted from
raw MSL notes with ai_query():
topic · info need · sentiment ·
interest · follow-up · competitor ·
evidence type · topic_category*

Unstructured → structured, LLM-powered

GOLD

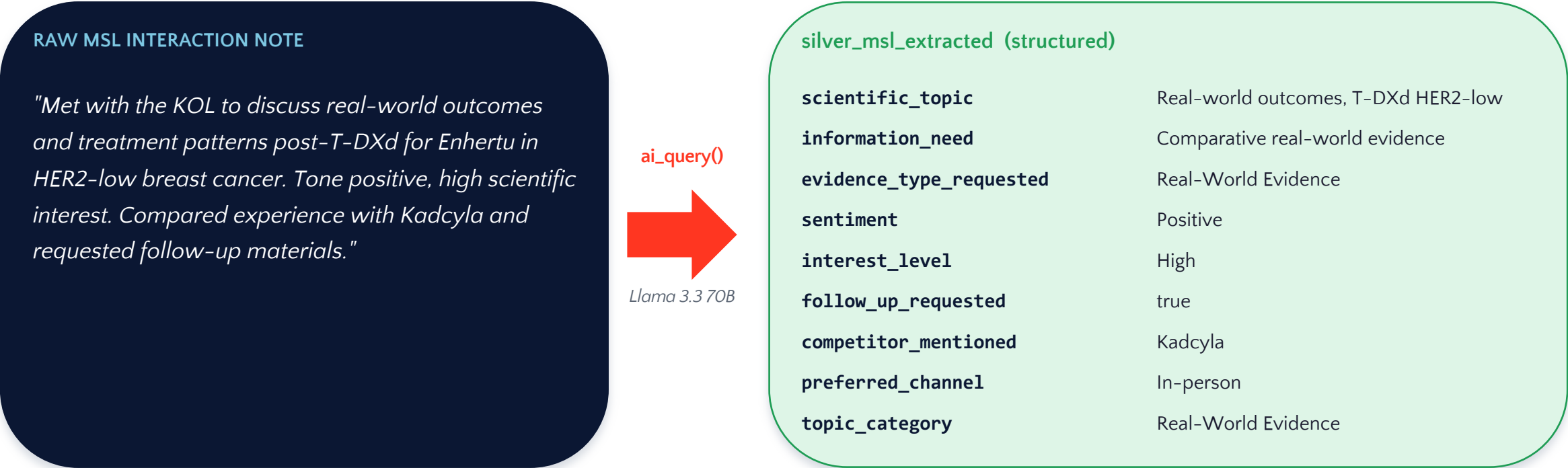
- gold_hcp_360
- gold_hcp_priority
- gold_medical_topic_interest
- gold_next_best_engagement
- gold_territory_opportunity
- gold_congress_engagement
- gold_approved_content /
_content_shared

Analytics-ready, aggregated & scored

31 Delta tables · 2 governed metric views · 2 UC Volumes (raw + source_documents) — Lakeflow-built, Lakebase-served, one lineage graph

From free-text notes to structured intelligence

The richest medical signal is locked in unstructured MSL notes. We unlock it in-place with `ai_query()` — no data movement, fully governed, at SQL scale.



One SQL function turns every MSL note into governed, queryable columns — the foundation for topic-interest scoring, prioritization, and next-best-engagement in Gold.

Genie space: ask Medical Affairs questions in English

"Medical Omnichannel Intelligence — Daiichi Sankyo" Genie space lets medical & field teams self-serve governed answers — no SQL, no dashboard-hunting.

Example questions

- Which 5 territories have the lowest 90-day HCP reach rate?
- Which high-priority KOLs are under-engaged and overdue for an MSL visit?
- Show medical inquiries overdue against SLA, by product.
- What are the top scientific topics of interest across oncology KOLs?
- List Tier 1 KOLs with open inquiries about Safety / ILD Management.

Why it matters

- Governed by Unity Catalog — answers respect the same permissions as everything else
- Runs on the curated Gold tables & metric views — consistent definitions
- Trusted metrics: KPIs come from metric_hcp_engagement & metric_medical_inquiries
- Democratizes access for non-technical medical stakeholders
- Transparent — shows the generated SQL behind every answer

AI/BI dashboard: the omnichannel command center

A curated, always-on view of Medical Affairs engagement — 6 KPIs, reach by territory, topic interest, inquiry SLAs, priority tiers and the next-best-engagement queue — built on the same governed Gold tables and metric views that power Genie and the app.

HCP 360

Unified profile,
engagement & interest

Priority & segments

MSL priority score,
under-engaged flags

Topic interest

Emerging scientific
topics over time

Inquiry SLAs

Response times &
overdue tracking

Territory opportunity

Reach rate &
opportunity score

Congress engagement

Sessions, booths,
distinct events

Same governed source as Genie & the Copilot — build a metric once, trust it everywhere.

Medical Engagement Copilot — the Databricks App

Select a KOL and get a pre-engagement briefing that unites analytics, AI, governed content, and a next-best action — the MSL's single prep screen. The action queue is served from Lakebase; review decisions are written back to Lakebase.

1	Select an HCP	Prioritized queue served from Lakebase — highest MSL priority score first.
2	Unified HCP 360	Profile, recent interactions, top scientific interests, inquiries, congress, publications, content already shared, open follow-ups.
3	AI briefing	3–4 sentence narrative via <code>ai_query()</code> , grounded ONLY in that HCP's structured context — factual, non-promotional.
4	Plan, approved content + Lakebase write-back	Objective, topic, channel, timing — plus an approved asset (with med review ID). The MSL logs the decision, written back transactionally to Lakebase. Requires MSL review.

Runs as a Databricks App (service-principal auth, no hardcoded tokens) reading governed Gold over serverless SQL and the Lakebase MSL action queue.

Impact in your KPIs — baseline → target

Reach

62% → 80%

Priority-KOL 90-day reach, lifted by under-engaged flags + priority score. -\$1.5M/yr.

Speed

-2h → mins

MSL prep per engagement — AI briefing replaces manual context assembly. -\$1.3M/yr.

Service

71% → 95%

Medical-inquiry SLA compliance — overdue items flagged by product. -\$0.6M/yr.

Insight

0 → 500+

Free-text MSL notes turned into queryable topic-interest signal.

The ROI, quantified

-\$3.4M / year estimated for a 120-MSL field-medical org — faster prep, sharper prioritization and faster SLAs — on one governed Lakehouse, no data copies, one security model.

Why this approach — and what we consciously traded



Lakebase for serving

The app needs single-row reads + transactional write-back — Delta Gold can't hit that latency. Trade-off: a small Postgres surface, synced from UC.



ai_query() in-pipeline

Extraction stays in governed SQL, no data egress. An external LLM API adds movement, cost & compliance risk. Trade-off: hosted models only.



Genie AND a dashboard

Genie for open-ended questions, AI/BI for always-on KPIs — same governed metrics. One alone under-serves either explorers or operators.



Llama 3.3 70B, hosted

In-workspace, no third-party data sharing, strong extraction. Trade-off: not the frontier model — ample for structured extraction.



One UC schema

A single medallion, one lineage graph & security model — vs. fragmented marts. Trade-off: tighter coordination on a shared schema.



Synthetic data

Safe, reproducible, no PHI/PII — vs. real data that blocks sharing. Trade-off: value figures are modeled estimates, not measured.

AI as a force multiplier — tools, prompts & patterns



Claude Code / Isaac

An agentic AI teammate drove the build end to end — provisioning, code and this deck — not just autocomplete.



Databricks MCP tools

UC, Lakeflow, Genie, Lakebase and Apps provisioned from natural-language intent — infrastructure as conversation.



Extraction prompt pattern

One reusable `ai_query()` prompt turns every free-text MSL note into 9 governed columns — the product's core AI move.



Grounded-briefing guardrail

The app's prompt enforces factual, context-only, non-promotional output — a safety pattern for regulated GenAI.



Generate → verify loop

Every resource was validated live (`execute_sql`, endpoint checks) before moving on — AI proposes, we verify.



Synthetic data generation

Faker + Spark patterns produced realistic KOL, inquiry and engagement data at demo scale.

Built for a regulated, non-promotional context



Human-in-the-loop

Every AI output is a recommendation for MSL review — never an automated action or outreach.



Only approved content

The app surfaces medically-approved assets only, each with a med_review_id, and never re-shares what an HCP already has.



Unity Catalog governance

One permission & lineage model across tables, metrics, Genie, dashboard, Lakebase and the app — auditable end to end; enforced UC tag policies, napi classified PII.



Grounded, non-promotional

Briefings are grounded ONLY in the HCP's structured context; prompt enforces factual, evidence-based, non-promotional tone.



Synthetic data

The entire demo runs on synthetic data — safe to show, no PII, no real patient or HCP information.



One source of truth

Governed metric views mean the same KPI definition powers every surface — no conflicting numbers.

Thank You